In a strongly connect component (*SCC*), every vertex is reachable from any other vertex; strongly connected components must therefore have cycles (see section 14.2.3).

The notion of strongly connected components is particularly important: it allows to define a graph of the SCCs, which is going to be sensibly smaller than the original graph, and run many algorithms on this graph instead: we can gain a great speedup by first examining a graph at a high level, and then (possibly) studying the interaction within each SCC.

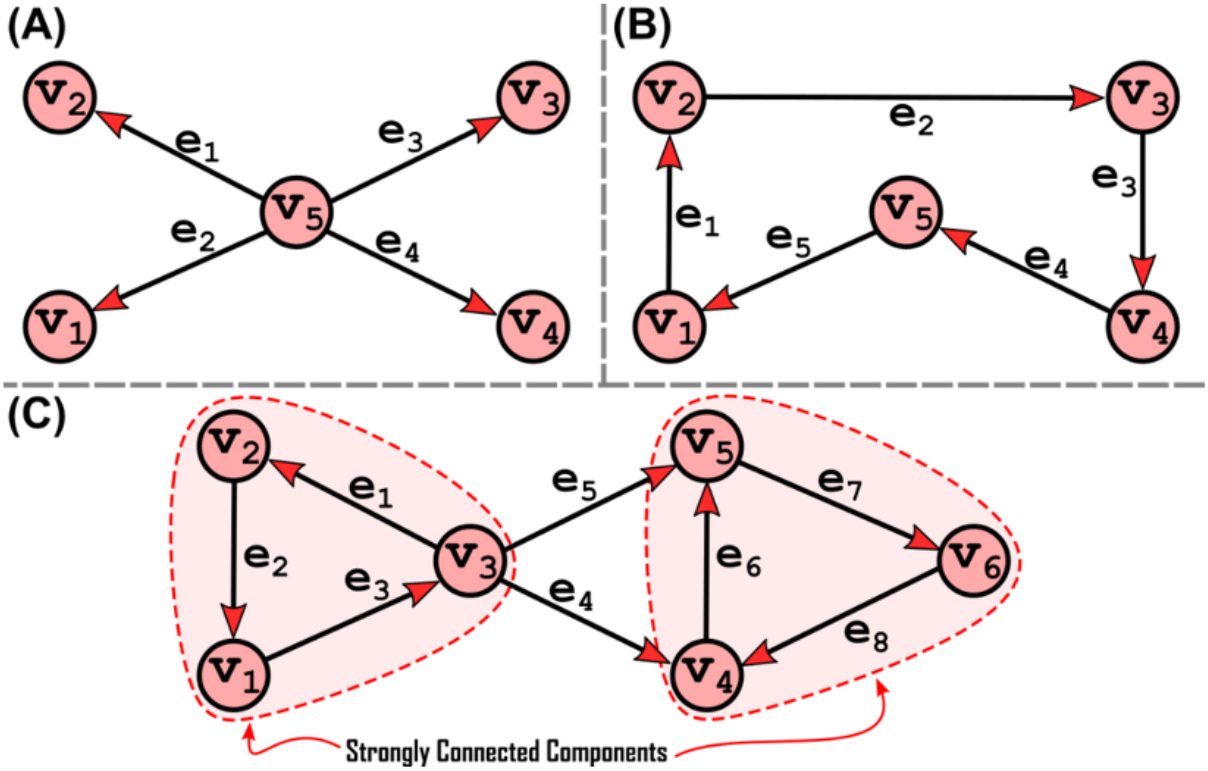


Figure 14.6 Only graph (B) is strongly connected, but graph (C) has two strongly connected components.

A tree is usually regarded as an undirected graph, at least in graph theory; but in implementations, usually, each node stores the links to its children, and only sometimes to its